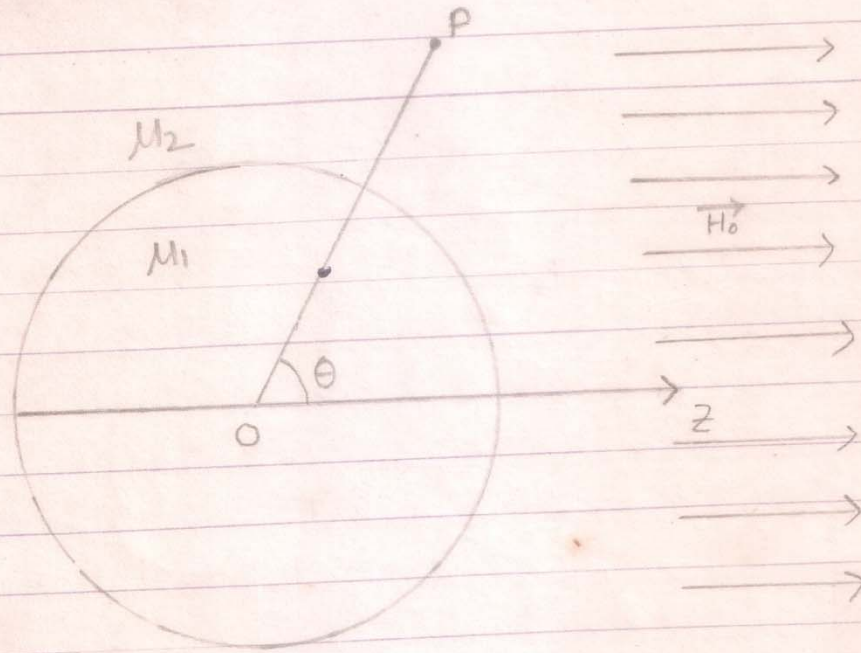




Uniformly magnetised sphere in external magnetic field:



Given: μ_1 & μ_2 relative permialubility inside and outside the sphere.

a = radius of the sphere.

$\vec{H}_1$ and $\vec{H}_0$ field inside & outside the sphere.

θ = angle between the direction of radius vector and the direction of magnetic field $\vec{H}_0$.

If ϕ_1 & ϕ_2 be the potential at a point $P(r, \theta)$ inside and outside the sphere then:



$$\phi_1 = -H_1 r \cos\theta \quad \dots (1) \quad \text{when } r < a$$

$$\phi_2 = -H_0 r \cos\theta + \frac{A \cos\theta}{r^2} \quad \dots (2) \quad \text{when } r > a$$

The term $\frac{A \cos\theta}{r^2}$ is absent in the expression for ϕ_1 because r^2 with its inclusion ϕ_1 will be infinity at $r=0$ i.e. at the centre of the sphere potential becomes infinity which is not possible.

Applying Boundary Condition:

Since there is no surface current tangential component of $\vec{H}$ is continuous across the boundary. Again as this boundary condition has been derived from Ampere's circuital law $\oint \vec{H} \cdot d\vec{l}$ which is a measure of potential this boundary condition is equivalent to that "the potential is continuous across the boundary".

$$\text{or } [\phi_1]_{r=a} = [\phi_2]_{r=a}$$

Putting eqⁿ (1) and (2):

$$-H_1 a \cos\theta = -H_0 a \cos\theta + \frac{A \cos\theta}{a^2}$$

$$\text{or } \boxed{H_1 = H_0 - \frac{A \cos\theta}{a^3}} \quad \dots (3) \quad \left| \quad (H_0 - H_1) a^3 = A \right.$$

$M' = \chi_m H_1 = (M_1 - 1) H_1$ = The magnetisation induced in the sphere due to the field inside the sphere.



M_0 = Permanent magnetisation of the Sphere.

$\therefore$ Total magnetisation of the Sphere

$$M_1 = M' + M_0 = (M_1 - 1) H_1 + M_0$$

$\therefore$ The induction vector inside the Sphere

$$\vec{B}_1 = \mu_0 (\vec{H}_1 + \vec{M}_1)$$

$$\text{or, } \vec{B}_1 = \mu_0 [\vec{H}_1 + (M_1 - 1) \vec{H}_1 + \vec{M}_0]$$

$$\text{or, } \vec{B}_1 = \mu_0 M_1 \vec{H}_1 + \mu_0 \vec{M}_0 \quad \text{--- (4)}$$

Again,

The magnetic induction vector outside the Sphere :

$$\vec{B}_2 = \mu_0 M_2 \vec{H}_0 \quad \text{--- (5)}$$

We now find radial component of induction vectors $\vec{B}_1$ and $\vec{B}_2$

The radial component of the induction vector inside the medium is $B_1 \cos \theta$

$$B_1 \cos \theta = \mu_0 M_1 H_1 \cos \theta + \mu_0 M_0 \cos \theta \quad \text{--- (6)}$$

The radial component of $\vec{B}_2$ outside the Sphere is $B_2 \cos \theta$

$$B_2 \cos \theta = -\mu_0 M_2 \frac{\partial \phi_2}{\partial r}$$

$$= -\mu_0 M_2 \frac{\partial}{\partial r} (-H_0 r \cos \theta + A r^{-2} \cos \theta)$$

$$B_2 \cos \theta = \mu_0 M_2 \left(\vec{H}_0 + \frac{2A}{r^3} \right) \cos \theta \quad \text{--- (7)}$$



Applying the boundary condition

$$(B_1)_\perp = (B_2)_\perp$$

$$\cancel{\mu_0} (M_1 H_1 + M_0) \cancel{\cos\theta} = \cancel{\mu_0} M_2 \left(H_0 + \frac{2A}{a^3} \right) \cancel{\cos\theta}$$

Putting the value of A:

$$M_1 H_1 + M_0 = M_2 (H_0 + 2H_0 - 2H_1) \quad | \text{ term (3)}$$

$$\text{or, } M_1 H_1 + 2M_2 H_1 = 3M_2 H_0 - M_0$$

$$\text{or, } H_1 = \frac{3M_2}{M_1 + 2M_2} H_0 - \frac{1}{M_1 + 2M_2} M_0$$

The difference $H_d = H_1 - H_0$ is called demagnetising field.